Medical Physics
The International Journal of Medical Physics Research and Practice

Parker weights revisited

Stefan Wesarg, Matthias Ebert, and Thomas Bortfeld

Citation: Medical Physics **29**, 372 (2002); doi: 10.1118/1.1450132
View online: http://dx.doi.org/10.1118/1.1450132
View Table of Contents: http://scitation.aip.org/content/aapm/journal/medphys/29/3?ver=pdfcov
Published by the American Association of Physicists in Medicine

Articles you may be interested in

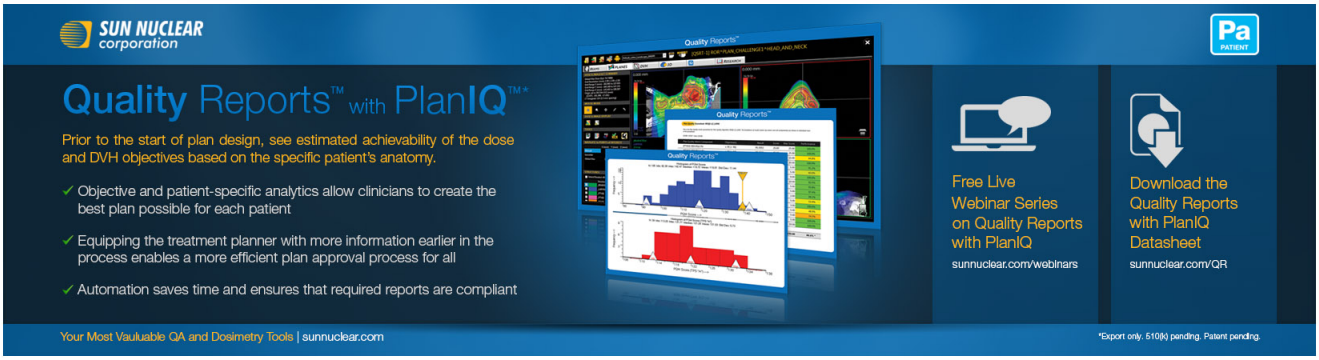
Noise-weighted spatial domain FBP algorithm
Med. Phys. **41**, 051906 (2014); 10.1118/1.4870989

Optimal "image-based" weighting for energy-resolved CT
Med. Phys. **36**, 3018 (2009); 10.1118/1.3148535

Regularized iterative weighted filtered backprojection for helical cone-beam CT
Med. Phys. **35**, 4173 (2008); 10.1118/1.2966353

Parker weights revisited
Med. Phys. **29**, 1916 (2002); 10.1118/1.1496098

Fast reconstruction with uniform noise properties in halfscan computed tomography
Med. Phys. **27**, 2031 (2000); 10.1118/1.1288239



Parker weights revisited

Stefan Wesarg[a)]
Fraunhofer Institute for Computer Graphics, Cognitive Computing and Medical Imaging (A7), Rundeturmstr. 6, 64283 Darmstadt, Germany

Matthias Ebert[b)]
MRC Systems GmbH, Hans-Bunte-Str. 10, 69123 Heidelberg, Germany

Thomas Bortfeld[c)]
Northeast Proton Therapy Center, 30 Fruit Street, Boston, Massachusetts 02114

(Received 23 April 2001; accepted for publication 15 October 2001; published 21 February 2002)

The short-scan case in fan-beam computed tomography requires the introduction of a weighting function to handle redundant data. Parker introduced such a weighting function for a scan over π plus the opening angle of the fan. In this article we derive a general class of weighting functions for arbitrary scan angles between π plus fan angle and 2π (over-scan). These weighting functions lead to mathematically exact reconstructions in the continuous case. Parker weights are a special case of a weighting function that belongs to this class. It will be shown that Parker weights are not generally the best choice in terms of noise reduction, especially when there is considerable over-scan. We derive a new weighting function that has a value of 0.5 for most of the redundant data and is smooth at the boundaries. © *2002 American Association of Physicists in Medicine.*
[DOI: 10.1118/1.1450132]

Key words: computed tomography (CT), fan-beam geometry, short-scan reconstruction, Parker weights

I. INTRODUCTION

In x-ray computed tomography (CT), objects can be reconstructed from projection data collected over an angular range of 2π. However, it is not necessary to acquire projections over the full circle to reconstruct a complete cross section of the object. Using so-called ''short-scan'' approaches, the scanning time is reduced and the total dose is potentially decreased. In third-generation fan-beam CT scanners, the range of gantry angles required to collect the ''minimal complete data set'' equals π plus the opening angle of the fan. However, a short-scan set of fan projections cannot be reconstructed by means of straightforward filtered backprojection because it contains redundant data. There are various approaches for considering the redundant data.

(i) In *parallel rebinning* algorithms as proposed by Dreike and Boyd,[1] a set of equivalent parallel-beam projections within a view angle range of π is obtained from the recorded short-scan fan-beam data. Except for specific acquisition geometries, this so-called rebinning step requires two successive (linear) interpolations for each line integral in the parallel-beam data set. Spatial resolution is therefore potentially slightly decreased. Once the parallel projections have been estimated, a standard parallel-beam image reconstruction algorithm is applied.

(ii) In *complementary rebinning* techniques, the missing part of the short-scan projection data set is filled out using the measured rays. Standard filtered backprojection is then applied. Complementary rebinning can be performed without interpolation on discrete data if the central ray is exactly centered with respect to the detector array (i.e., if no detector quarter shift for virtually doubling the sampling frequency is applied) and if the view angle pitch is twice the fan angle pitch. In practice, CT scanners take substantially fewer projections per turn than suggested by the latter condition.

(iii) Naparstek[2] showed that the problem of redundant data can be solved by applying a suitable *weighting function* to the sinogram. Primarily, he used a sharp binary weighting function, but he also mentioned that improved results could be achieved by using smoother functions. Parker[3] introduced a smooth weighting function applicable to the minimal complete data set. These weights have become known as the ''Parker weights'' and have been widely used since then. The Parker weights were recently extended to the case of data sets larger than the minimal complete one, and less than or equal to that of a 2π scan, which is often referred to as ''over-scan.''[4] A step towards generalized smooth weighting functions was made by Crawford and King.[5]

(iv) Pan[6] proposed a *hybrid technique* for handling short-scan fan-beam projections. This method consists of sinogram weighting such as described above, calculation of the Fourier series expansion of the parallel-beam sinogram from the weighted fan-beam sinogram, and standard parallel-beam image reconstruction. Images reconstructed by the hybrid technique exhibit lower and spatially more uniform noise than those reconstructed by the pure weighting technique.

This article deals with sinogram weighting. To the knowledge of the authors, there is no proof in the literature that the Parker weights lead to accurate results or that they are mathematically exact. In fact, Zeng and Gullberg[7] argue that a short-scan reconstruction involving Parker weights would be an approximation, but they do not give a mathematical proof either.

In this article, we derive a sufficient condition for weighting functions that guarantees mathematically exact reconstructions for filtered backprojection based on continuous sinograms. These weighting functions are applicable to minimal complete as well as to over-scan data sets. We show that the Parker weights belong to this general set of functions. Furthermore, we derive a specific function that considers the discretization of the problem and the effect of photon noise in the projections. Finally, we demonstrate that, in the presence of noise, this new weighting function leads to slightly better results than the simple extension of the Parker weights previously proposed by Silver.[4]

II. SHORT-SCAN WEIGHTS FOR CONTINUOUS DATA

To investigate feasible short-scan weights for continuous data, the well-known reconstruction formula for parallel projections is first transformed to fan-beam geometry. At this point, a general weighting function is introduced. A sufficient condition for the weights is then derived which ensures mathematically exact reconstruction.

A. Formulation of the problem

In the case of a parallel-beam geometry, the reconstruction formula is given by

$$f(\mathbf{r})=\int_{-\infty}^{\infty}\int_{\delta}^{\pi+\delta} g(\phi,t)h(\mathbf{r}\cdot\hat{\mathbf{n}}_{\phi}-t)\,\mathrm{d}\phi\,\mathrm{d}t, \tag{1}$$

where $g(\phi,t)$ is the line integral of $f(\mathbf{r})$ at view angle ϕ and detector position t, $\mathbf{r}$ is the point to be reconstructed, and $\hat{\mathbf{n}}_{\phi}$ is the unit vector perpendicular to the projection ray. Usually, the ϕ-integral is taken over the interval from 0 to π. For reasons that will soon become apparent, we introduce a shift by δ. The function h is a reconstruction filter such as the kernel proposed by Ramachandran and Lakshminarayanan (Ram–Lak filter) or the one by Shepp and Logan; see, for example, Ref. 8. Using the Ram–Lak filter, Eq. (1) yields an exact reconstruction for band-limited functions.

The parallel-beam coordinates (ϕ,t) are now replaced by fan coordinates (β,α), where β is the view angle of the fan projection, and α is the angle within the fan. Provided the object to be reconstructed is of limited spatial extent, the fan angle α can be restricted to a certain range from $-\delta$ to δ with $\delta<\pi/2$. Hence, the opening angle of the fan, which is assumed to cover the object completely in all views, equals 2δ. The relationship between parallel and fan coordinates is $t=R\sin\alpha$ and $\phi=\beta+\alpha$ with the Jacobian $R\cos\alpha$. The reader is referred to Ref. 8 for further details.

The reconstruction formula for fan-beam data sets can now be written as

$$f(\mathbf{r})=\int_{-\delta}^{\delta}\int_{\delta-\alpha}^{\pi+\delta-\alpha} g(\beta,\alpha)R\cos\alpha \times h(\mathbf{r}\cdot\hat{\mathbf{n}}_{\beta+\alpha}-R\sin\alpha)\,\mathrm{d}\beta\,\mathrm{d}\alpha. \tag{2}$$

This formula cannot be implemented as filtered backprojection because the integration over the view angle β depends on the fan angle α. The order of integration, therefore, can not be changed such that the filtering step is performed before and independent of the backprojection. However, if the integration range within the sinogram depicted in Fig. 1(a) is extended from the parallelogram III to the rectangle I+II+III, the β integration becomes independent of α.

The projection rays which are represented by the two triangles I and II are redundant because the middle region III already contains that data. By extending the integration to the whole rectangle, the introduction of a weighting function is necessary to handle the extra (redundant) data. Denoting this weight by $\omega(\beta,\alpha)$, the reconstruction formula becomes

$$f(\mathbf{r})=\int_{-\delta}^{\delta}\int_{0}^{\pi+2\delta} \omega(\beta,\alpha)g(\beta,\alpha)R\cos\alpha \times h(\mathbf{r}\cdot\hat{\mathbf{n}}_{\beta+\alpha}-R\sin\alpha)\,\mathrm{d}\beta\,\mathrm{d}\alpha. \tag{3}$$

In this equation, the order of integration can be changed. The only difference between this short-scan and the full-scan formula is, besides the shorter range of β integration, the weighting function $\omega(\beta,\alpha)$ which has to be applied before reconstruction. In particular, Eq. (3) can be rewritten in the form of a modified filtered backprojection algorithm as shown, for example, in Ref. 8.

With the following definition of a binary weighting function,

$$\omega_1(\beta,\alpha)=\begin{cases}1 & \text{if } \delta-\alpha\leqslant\beta\leqslant\pi+\delta-\alpha,\\ 0 & \text{otherwise,}\end{cases} \tag{4}$$

Eq. (3) becomes equivalent to Eq. (2). However, in practical reconstruction algorithms operating on discrete data the integrals have to be approximated by sums. In this case, the application of $\omega_1(\beta,\alpha)$ leads to large numerical errors because of its discontinuities, as already shown by Naparstek.[2] Fortunately, $\omega_1(\beta,\alpha)$ is not the only possible weighting function. A sufficient condition for weighting functions which ensures mathematically exact reconstructions in the continuous case is derived in the following. Weighting functions which satisfy this condition and are applicable to discrete data are also investigated.

B. Symmetries in fan-beam geometry

The integrand in Eq. (3) satisfies some symmetry conditions which are needed for the following derivation. First of all, the line integral for ray (β,α) equals that of ray $(\beta+\pi+2\alpha,-\alpha)$ as shown in Ref. 8. Furthermore, the vector $\hat{\mathbf{n}}_{\beta+\alpha}$ is reversed if the index $\beta+\alpha$ is changed to $\beta+\pi+\alpha$, i.e., $\hat{\mathbf{n}}_{\beta+\alpha}=-\hat{\mathbf{n}}_{\beta+\pi+\alpha}$. The filter kernel h exhibits the symmetry

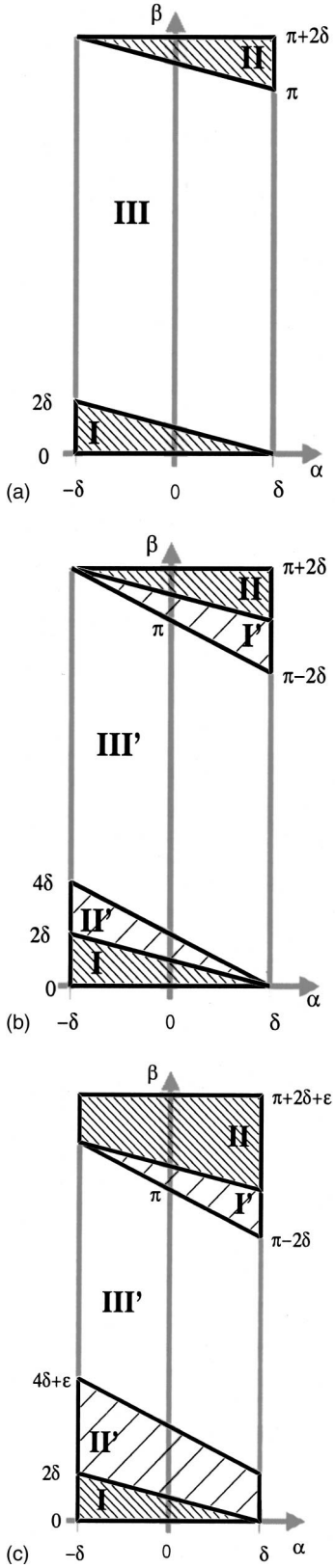


FIG. 1. Short-scan sinograms. (a) Visualization of integration range III according to Eq. (3) and redundant data within triangles I and II if the parallelogram is extended to a rectangle, (b) "Parker sinogram," and (c) extended short-scan sinogram considering an over-scan ϵ

$h(t)=h(-t)$. Hence, a symmetry condition for the integrand of Eq. (3), which is denoted as $\tilde{g}(\beta,\alpha)$ in the following, is given by

$$\begin{aligned}\tilde{g}(\beta,\alpha)&=g(\beta,\alpha)R\cos\alpha h(\mathbf{r}\cdot\hat{\mathbf{n}}_{\beta+\alpha}-R\sin\alpha)\\&=g(\beta+\pi+2\alpha,-\alpha)R\cos(-\alpha)\\&\quad\times h(-\mathbf{r}\cdot\hat{\mathbf{n}}_{\beta+\alpha}+R\sin\alpha)\\&=\tilde{g}(\beta+\pi+2\alpha,-\alpha).\end{aligned}\tag{5}$$

Using this symmetry, a condition for the weight $\omega(\beta,\alpha)$ is derived in the following subsection.

C. Derivation of short-scan weights

The equality $g(\beta,\alpha)=g(\beta+\pi+2\alpha,-\alpha)$ of two opposite rays leads to a modified sinogram which is often referred to as "Parker sinogram" [3] [see Fig. 1(b)]. The regions I′ and II′ are part of the former region III. These two regions contain data that are also contained in regions I and II, respectively, which is evident from the symmetry mentioned above.

A weighting function $\omega(\beta,\alpha)$ leads to a mathematically exact reconstruction if the integral in Eq. (3) equals those of the full-scan case given in Eq. (2). Obviously, the weighting function has to be one within the parallelogram III′ because there is no redundancy in this region. Therefore, the integration over the two regions I′ and II′ without applying a weighting function has to equal the integration over the four regions I, I′, II, and II′ involving the weights $\omega(\beta,\alpha)$. This is the only necessary condition for $\omega(\beta,\alpha)$ required in the continuous case.

Before deriving a set of weighting functions, the "Parker sinogram" is extended to data sets that are larger than the minimal complete one. Scanning over a range greater than $\pi+2\delta$ leads to an "over-scan" ϵ, $0\leqslant\epsilon\leqslant\pi-2\delta$. This results in the sinogram shown in Fig. 1(c). For $\epsilon>0$, a rectangle is added to region II and a parallelogram is added to region II′, which are then both trapezoid-shaped. Including the over-scan, Eq. (3) is replaced by

$$\begin{aligned}f(\mathbf{r})=\int_{-\delta}^{\delta}\int_{0}^{\pi+2\delta+\epsilon}&\omega(\beta,\alpha)g(\beta,\alpha)R\cos\alpha\\&\times h(\mathbf{r}\cdot\hat{\mathbf{n}}_{\beta+\alpha}-R\sin\alpha)\,\mathrm{d}\beta\,\mathrm{d}\alpha,\end{aligned}\tag{6}$$

assuming that the definition of $\omega(\beta,\alpha)$accounts for a specific over-scan ϵ.

The corresponding regions in Fig. 1(c) to be considered are region I with region I′ and region II with region II′. For each of them, the integral containing the weight $\omega(\beta,\alpha)$ is compared with the unweighted integral over I and II′, respectively.

Starting with regions I and I′, the symmetry derived above leads to

$$\int_{-\delta}^{\delta}\int_{0}^{\delta-\alpha}\omega(\beta,\alpha)\tilde{g}(\beta,\alpha)\,\mathrm{d}\beta\,\mathrm{d}\alpha + \int_{-\delta}^{\delta}\int_{\pi-2\alpha}^{\pi+\delta-\alpha}\omega(\beta,\alpha)\tilde{g}(\beta,\alpha)\,\mathrm{d}\beta\,\mathrm{d}\alpha$$

$$= \int_{-\delta}^{\delta}\int_{0}^{\delta-\alpha}\omega(\beta,\alpha)\tilde{g}(\beta,\alpha)\,\mathrm{d}\beta\,\mathrm{d}\alpha - \int_{\delta}^{-\delta}\int_{0}^{\delta-\alpha}\omega(\beta+\pi+2\alpha,-\alpha) \times\tilde{g}(\beta+\pi+2\alpha,-\alpha)\,\mathrm{d}\beta\,\mathrm{d}\alpha \tag{7}$$

$$= \int_{-\delta}^{\delta}\int_{0}^{\delta-\alpha}[\omega(\beta,\alpha)+\omega(\beta+\pi+2\alpha,-\alpha)] \times\tilde{g}(\beta,\alpha)\,\mathrm{d}\beta\,\mathrm{d}\alpha. \tag{8}$$

If the sum of the weights in the rectangular brackets is 1, the last integral is the integral of $\tilde{g}(\beta,\alpha)$ over region I. This equals the integral of $\tilde{g}(\beta,\alpha)$ over region I′ thus showing the condition for exactness is fulfilled.

Accordingly, for regions II and II′ the integration is given by

$$\int_{-\delta}^{\delta}\int_{\pi+\delta-\alpha}^{\pi+2\delta+\epsilon}\omega(\beta,\alpha)\tilde{g}(\beta,\alpha)\,\mathrm{d}\beta\,\mathrm{d}\alpha + \int_{-\delta}^{\delta}\int_{\delta-\alpha}^{2\delta-2\alpha+\epsilon}\omega(\beta,\alpha)\tilde{g}(\beta,\alpha)\,\mathrm{d}\beta\,\mathrm{d}\alpha$$

$$= -\int_{\delta}^{-\delta}\int_{\delta-\alpha}^{2\delta-2\alpha+\epsilon}\omega(\beta+\pi+2\alpha,-\alpha) \times\tilde{g}(\beta+\pi+2\alpha,-\alpha)\,\mathrm{d}\beta\,\mathrm{d}\alpha + \int_{-\delta}^{\delta}\int_{\delta-\alpha}^{2\delta-2\alpha+\epsilon}\omega(\beta,\alpha)\tilde{g}(\beta,\alpha)\,\mathrm{d}\beta\,\mathrm{d}\alpha \tag{9}$$

$$= \int_{-\delta}^{\delta}\int_{\delta-\alpha}^{2\delta-2\alpha+\epsilon}[\omega(\beta+\pi+2\alpha,-\alpha)+\omega(\beta,\alpha)] \times\tilde{g}(\beta,\alpha)\,\mathrm{d}\beta\,\mathrm{d}\alpha. \tag{10}$$

If the sum of the weights in the rectangular brackets equals 1 again, the last integral is the integral of $\tilde{g}(\beta,\alpha)$ over region II′.

Hence, in both cases the desired equality of the integrals can be satisfied by the condition

$$\omega(\beta,\alpha)+\omega(\beta+\pi+2\alpha,-\alpha)=1. \tag{11}$$

As pointed out earlier, $\omega(\beta,\alpha)$ must have the value of unity within the middle region III′, since this area does not contain any redundant data. Introducing a function $\rho(\beta,\alpha)$, $\rho\in[0,1]$, defined in the lower trapezoid of the sinogram, and using Eq. (11), the desired function $\omega(\beta,\alpha)$ can be written in the following form:

$$\omega(\beta,\alpha)=\begin{cases}\rho(\beta,\alpha) & \text{if } 0\leqslant\beta\leqslant 2\delta-2\alpha+\epsilon,\\ 1 & \text{if } 2\delta-2\alpha+\epsilon<\beta<\pi-2\alpha,\\ 1-\rho(\beta-\pi+2\alpha,-\alpha) & \text{if } \pi-2\alpha\leqslant\beta\leqslant\pi+2\delta+\epsilon.\end{cases} \tag{12}$$

This formula describes a sufficient condition for short-scan weights which, applied prior to continuous filtered backprojection, ensures mathematically exact results.

III. SHORT-SCAN WEIGHTS FOR DISCRETE DATA

Although the sufficient condition for short-scan weights derived in the previous section is fairly simple, the situation is more complicated for discrete and noisy data. This is outlined in the following subsection. The Parker weights,[3] which have been widely utilized in short-scan implementations, are then revisited using a new notation. Furthermore, an alternative weighting function is defined and compared to the Parker weights based on computer simulations.

A. Design requirements

There are two conflicting requirements when appropriate short-scan weights are to be defined which consider both the discrete and noisy character of projection data obtained in real measurements.

On one hand, the weighting function $\omega(\beta,\alpha)$ should not introduce aliasing artifacts. Therefore, $\omega(\beta,\alpha)$ has to be band-limited ideally. This cannot be achieved in practical implementations, since a band-limited function does not exhibit compact support. In the literature, short-scan weights are often required to be a smooth function in a quite general sense. This means that its two-dimensional Fourier transform with respect to β and α should vanish above the Nyquist frequencies determined by the sampling intervals in order to avoid severe aliasing.

On the other hand, a high signal-to-noise ratio (SNR) in the images is desired when reconstructions are performed on measured projection data. In the following, only the photon statistics of the x-ray transmission data are considered; other sources of noise such as, for example, electronic noise in the detector are neglected. A short-scan weighting function $\omega_{0.5}(\beta,\alpha)$ applying a value of 0.5 to all redundant data is optimal in terms of noise reduction, since the measurements of all line integrals can be assumed to be statistically independent. A rigorous mathematical proof of this statement follows directly from the Gaussian law of error propagation and is therefore not included here. The function $\omega_{0.5}(\beta,\alpha)$, however, introduces infinitely high frequencies into the sinogram.

It is obvious that there must be a trade-off between these two aims. It is therefore not possible to define a general optimal short-scan weighting function. The, in some sense, best choice depends on the particular projection data set and the requirements of each specific application.

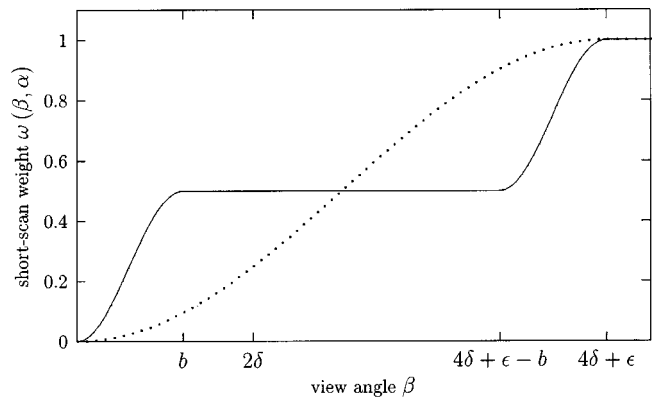


FIG. 2. Graph of the new weights (solid line) in comparison with the Parker weights (dotted line). The functions are plotted for $\alpha=-\delta$, $\epsilon=2\delta$, and (for the new weights) $q=0.25$. The new weighting function consists of two S-shaped parts of length $b=q\ (2\delta-2\alpha+\epsilon)$ at the sides to assure smoothness and a middle part with a constant value of 0.5.

B. Parker weights

One possible smooth form of $\omega(\beta,\alpha)$ is given by the Parker weights.[3] They can be composed of smooth S-shaped functions of the form

$$S(\beta'):=\begin{cases} 0 & \text{if } \beta'\leqslant-\frac{1}{2}, \\ \frac{1}{2}(1+\sin(\pi\beta')) & \text{if } |\beta'|<\frac{1}{2}, \\ 1 & \text{if } \beta'\geqslant\frac{1}{2}. \end{cases} \tag{13}$$

It is interesting to note that these functions result from a convolution of the Heaviside step function with the smoothing filter $(\pi/2)\cos(\pi\beta')$ within $|\beta'|\leqslant 1/2$. Of course, this is not the only possible smoothing filter. In fact, other filters might lead to better results, but this question is not within the scope of this article.

Adapting the S-functions to the lower trapezoid of Fig. 1(c) in the direction of β yields

$$\rho_{\mathrm{P}}(\beta,\alpha)=S\left(\frac{\beta}{2\delta-2\alpha+\epsilon}-\frac{1}{2}\right). \tag{14}$$

With the help of Eq. (12), the Parker weights $\omega_{\mathrm{P}}(\beta,\alpha)$ are obtained over the entire sinogram. They can be written as

$$\omega_{\mathrm{P}}(\beta,\alpha)=S\left(\frac{\beta}{2\delta-2\alpha+\epsilon}-\frac{1}{2}\right)-S\left(\frac{\beta-\pi+2\alpha}{2\delta+2\alpha+\epsilon}-\frac{1}{2}\right). \tag{15}$$

The Parker weights can be considered as a smoothed version of the binary weighting $\omega_1(\beta,\alpha)$ function in Eq. (4) obtained by filtering $\omega_1(\beta,\alpha)$ with a filter of spatially varying width in the direction of β.

Although the Parker weights are generally smooth, they still exhibit discontinuities at $(\beta,\alpha)=(0,\delta)$ and $(\beta,\alpha)=(\pi+2\delta,-\delta)$ if $\epsilon=0$. These discontinuities do not cause artifacts if the field of view is slightly larger than the object to be reconstructed, i.e., if $g(0,\delta)=g(\pi+2\delta,-\delta)=0$. Alternatively, practical difficulties can be avoided by introducing a small over-scan $\epsilon>0$.

C. A new weighting function

In this subsection, a new weighting function is to be developed which combines the advantageous property of $\omega_{0.5}(\beta,\alpha)$ in terms of high SNR with the smoothness of Parker weights to avoid streak artifacts. The weighting function should have a value of 0.5 over as wide an area as possible and smooth transition zones of length b at the boundaries.

An example of such a function is shown in Fig. 2. The length of the transition zones, b, can be set to a constant b_0. However, this requires some over-scan $\epsilon\geqslant 2b_0$ because otherwise there would not be enough space for the transition zone at the right side of the lower trapezoid and at the left side of the upper trapezoid in Fig. 1(c). Another possibility is to set b to a certain percentage q of the height B of the trapezoids in the direction of β. Let us consider the lower trapezoid. For a given α its height is $B(\alpha)=2\delta-2\alpha+\epsilon$. Thus, in this case b also depends on α, and it is $b(\alpha)=qB(\alpha)$.

A straightforward definition of the new weighting function ρ_{N} in the lower trapezoid is the sum of two S-shaped functions defined in Eq. (13) at two different positions which leads to

$$\rho_{\mathrm{N}}(\beta,\alpha)=\frac{1}{2}\left[S\left(\frac{\beta}{b(\alpha)}-\frac{1}{2}\right)+S\left(\frac{\beta-B(\alpha)}{b(\alpha)}+\frac{1}{2}\right)\right]. \tag{16}$$

Note that in the special case where $q=1$, i.e., $b(\alpha)=B(\alpha)$, these weights reduce to the Parker weights. With the help of the relation (12), the new weights $\omega_{\mathrm{N}}(\beta,\alpha)$ are determined in the whole sinogram. They can be written in the explicit form

$$\begin{aligned}\omega_{\mathrm{N}}(\beta,\alpha)=\frac{1}{2}\Bigg[&S\left(\frac{\beta}{b(\alpha)}-\frac{1}{2}\right)+S\left(\frac{\beta-2\delta+2\alpha-\epsilon}{b(\alpha)}+\frac{1}{2}\right)\\ &-S\left(\frac{\beta-\pi+2\alpha}{b(-\alpha)}-\frac{1}{2}\right)\\ &-S\left(\frac{\beta-\pi-2\delta-\epsilon}{b(-\alpha)}+\frac{1}{2}\right)\Bigg].\end{aligned} \tag{17}$$

D. Experimental comparison

The SNR results achievable with $\omega_{\mathrm{P}}(\beta,\alpha)$ and $\omega_{\mathrm{N}}(\beta,\alpha)$ were compared using synthetic projections which consider the photon statistics of x-ray transmission data. Beam-hardening, scatter, and electronic noise in the detector were neglected. A monoenergetic photon beam of 60 keV was assumed which is the mean energy in a typical, diagnostic 120 kVp spectrum.

The geometry of a typical third-generation CT scanner was used for the simulations. In this geometry, 1056 equally spaced fan-beam projections are collected over the full circle. The distance from the source to the axis of rotation is 57 cm, and the source-to-detector distance is 100 cm. The curved detector consists of 768 bins and covers a fan angle of ±26°. Each fan projection comprises 1536 samples at a spacing of half of the bin size to fulfill the Nyquist condition.

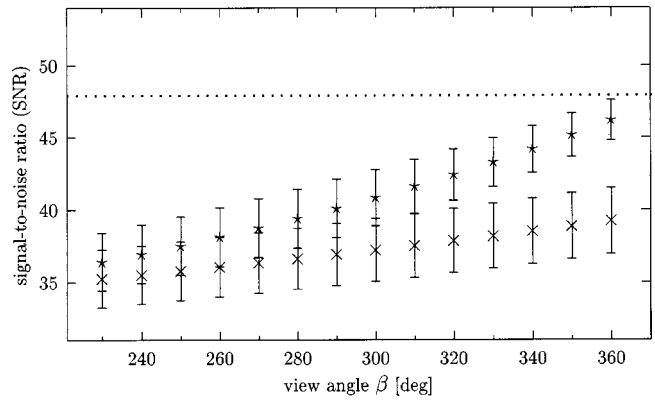


FIG. 3. SNR values obtained by using Parker weights (crosses) and the new weights (stars). The dotted, horizontal line indicates the SNR achievable in a full scan without applying any preweighting. The error bars represent the single standard deviation of the SNR values calculated in regions of interest, each of them comprising 128×128 pixels.

X-ray projections were simulated for a homogeneous water disc of 10 cm radius and an air calibration scan. At the energy of 60 keV, the linear attenuation coefficients for water and air are 2.046×10^{-1} cm^{-1} and 2.243×10^{-4} cm^{-1}, respectively.[9] For each detector sample, seven equally spaced line integrals, calculated analytically, were averaged in order to approximate stripe integration over the finite bin size. The line integrals were converted to the mean number of photons that is expected to be detected, assuming an effective source fluence of 6×10^5 photons per bin. The actual value for each sample in the phantom sinogram, N, and the air calibration sinogram, N_0, was drawn from a Poisson distribution with the corresponding mean detected signal. The final sinogram was calculated performing the common $\ln(N_0/N)$ normalization.

Using the parameters mentioned above and assuming a slice thickness of 3 mm, the complete CT scan would result in an effective dose of about 5 mSv at the center of the water phantom according to the conversion factors provided by Rogers.[10] This is typical for doses delivered during diagnostic x-ray CT examinations.

Ten CT images were reconstructed for the full-scan (without applying a weighting function) as well as for different short-scans involving extended Parker weights[3,4] and the new function derived above. The over-scan ϵ was varied between 0 (minimal complete data set) and $2\pi-2\delta$ (full data set). For the new weighting function, the parameter q was chosen to be 0.1, since this value has been found to be the smallest possible q avoiding streak artifacts for the geometry described above. The reconstruction algorithm utilizes a cotangent filter kernel applied in real space and a voxel-driven, weighted backprojection (see, e.g., Ref. 8 for further details). The images consist of 256×256 square pixels at a pitch of 0.5 mm in each direction. The signal-to-noise ratio was calculated in regions of interest comprising 128×128 pixels in ten images for each case.

The results of the computer simulations plotted in Fig. 3 indicate clearly that the new weighting function results in slightly higher SNR values than the Parker weights, especially if there is a considerable over-scan ϵ.

IV. DISCUSSION AND CONCLUSION

In this article, we derived a sufficient condition for short-scan weights to be used in fan-beam computed tomography. Any weighting function that fulfills this condition leads to an exact filtered backprojection reconstruction for ideal, i.e., continuous and noise-free, projection data, provided the cut-off frequency of the reconstruction filter is sufficiently high. The condition applies to minimal complete scanning as well as to overscan cases. The Parker weights and their recent extension to over-scan cases fulfill the condition and are therefore an example of a weighting function that is exact in the above sense, i.e., when applied to ideal data.

In the practically relevant discrete case, the weighting function must also be band-limited, such that the sampling condition can be fulfilled. In particular, the weights have to be smooth. The Parker weights are smooth but not band-limited. In fact, such band limitation directly conflicts with the requirement that the weighting function has to be of limited spatial extent. Hence, there is no way to make short-scan reconstruction with filtered backprojection exact when the projection data are discrete. As a result, there will always be more or less severe artifacts, generally showing up as streak artifacts. However, with the Parker weights these artifacts can be kept within an acceptable limit. We have shown that the S-shaped Parker weights can be considered as a convolution of a step function with a (cosine) smoothing filter. The symmetry condition could also be fulfilled by choosing any other spatially limited smoothing filter. Hence, it might be possible to achieve even better results with a filter that suppresses high spatial frequencies more effectively.

Another source of error besides discretization is noise in the projection data. For optimum noise reduction, the short scan weights should have a value of 0.5 for all redundant data and a value of unity for the nonredundant data. However, this sharp weighting function does not fulfill the band-limitation constraint. We therefore introduced a new weighting function that is a combination of the 0.5 weighting and the S-shaped Parker weights. Its application to artificial data shows that it leads to slightly improved signal-to-noise ratios when compared with the Parker weights. The larger the over-scan the more significant is this effect. The improvement is basically at no cost. As a further development it might be worthwhile to optimize the ratio between the length of the S-shaped transition zones and the 0.5-weighted region with respect to finding the best compromise between reduction of noise and avoidance of streak artifacts.

Although the derivations presented here consider only two-dimensional fan-beam CT, they can be generalized to a three-dimensional cone-beam geometry with a single-circular source trajectory using the Feldkamp approximation [11] in a straightforward manner.

ACKNOWLEDGMENT

We would like to thank Dr. M. Partridge for his careful proof-reading of the manuscript and his helpful comments.

[a]Electronic mail: stefan.wesarg@igd.fhg.de
[b]Electronic mail: m.ebert@mrc-heidelberg.de
[c]Electronic mail: tbortfeld@partners.org

[1] P. Dreike and D. B. Boyd, ''Convolution reconstruction of fan beam projections,'' Comput. Graph. Image Process. **5**, 459–469 (1976).
[2] A. Naparstek, ''Short-scan fan-beam algorithms for CT,'' IEEE Trans. Nucl. Sci. **NS-27**, 1112–1120 (1980).
[3] D. L. Parker, ''Optimal short scan convolution reconstruction for fan-beam CT,'' Med. Phys. **9**, 254–257 (1982).
[4] M. D. Silver, ''A method for including redundant data in computed tomography,'' Med. Phys. **27**, 773–774 (2000).
[5] C. R. Crawford and K. F. King, ''Computed tomography scanning with simultaneous patient translation,'' Med. Phys. **17**, 967–982 (1990).
[6] X. Pan, ''Fast reconstruction with uniform noise properties in halfscan computed tomography,'' Med. Phys. **27**, 2031–2036 (2000).
[7] G. L. Zeng and G. T. Gullberg, ''Short-scan cone beam algorithm for circular and noncircular detector orbits,'' Proc. SPIE **1233**, 453–463 (1990).
[8] A. C. Kak and M. Slaney, *Principles of Computerized Tomographic Imaging* (IEEE, New York, 1999).
[9] H. E. Johns and J. R. Cunningham, *The Physics of Radiology* (Thomas, Springfield, IL, 1983).
[10] D. W. O. Rogers, ''Fluence to dose equivalent conversion factors calculated with EGS3 for electrons from 100 keV to 20 GeV and photons from 11 keV to 20 GeV,'' Health Phys. **46**, 891–914 (1984).
[11] L. A. Feldkamp, L. C. Davis, and J. W. Kress, ''Practical cone-beam algorithm,'' J. Opt. Soc. Am. **1**, 612–619 (1984).